

Project Report: Silver Commodity Price Forecasting

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1. Executive Summary

This project presents an end-to-end data analytics and machine learning solution for silver commodity market forecasting. The platform integrates historical daily OHLCV market data with macroeconomic indicators and futures market term structures, providing a holistic view of price drivers. By employing ensemble learning techniques and visualizing results via a Power BI dashboard, the solution enables advanced trend analysis, correlation identification, and risk backtesting across critical historical economic crises.

2. Project Objectives

- Develop predictive models for daily silver spot price movements.
- Integrate multi-frequency datasets (Daily, Monthly, Futures).
- Construct a comprehensive business intelligence dashboard for decision support.

3. Data Architecture and Methodology

3.1 Data Sources

- **Silver OHLCV Data:** Historical daily spot prices, volume, and returns (2000–2025).
- **Macroeconomic Data:** Monthly indicators including the US Dollar Index (DXY) and gold prices.
- **Futures Data:** Contract months, futures prices, and basis calculations.

3.2 Data Preprocessing

Data cleaning involved addressing missing values, normalizing time-series structures, and establishing a robust Star Schema within Power BI to ensure efficient filtering and relationship modeling.

4. Machine Learning Pipeline

4.1 Feature Engineering

The Python pipeline (Pandas/Scikit-Learn) focused on generating technical indicators, calculating daily returns, and creating lagged features to improve model signal detection. Feature selection emphasized variables that correlate with silver's price volatility.

4.2 Model Training

An ensemble modeling approach was implemented, utilizing Random Forest and XGBoost algorithms. These models were evaluated for their ability to predict directional market moves. The output, `Ensemble_Pred`, provides a synthesized forecast integrated back into the Power BI ecosystem.

5. Power BI Dashboard Design

5.1 Executive Overview

The overview tab provides a high-level command center, featuring interactive KPI cards for latest spot prices and returns, coupled with time-series line charts that visualize the historical price trajectory from 2000 through the 2011 peak and the 2020–2025 market environment.

5.2 Macro & Futures Intelligence

This tab analyzes the external drivers of silver pricing. Correlation scatter plots compare silver with gold and the DXY index, while ribbon charts visualize the term structure of futures contracts, helping to identify contango and backwardation signals.

6. Crisis Backtesting

A custom `Crisis_Period` dimension was created via DAX to segment data based on major economic shocks, including the 2008 Financial Crisis and the 2020 COVID-19 pandemic. This allows users to benchmark cumulative returns and volatility during high-stress market events.

7. Technical Implementation

The integration of Python scripts for predictive modeling with the analytical visualization capabilities of Power BI provides an institutional-grade tool for commodity risk management.

8. Key Insights & Findings

Macroeconomic Influence: Analysis confirms a strong inverse relationship between silver performance and the DXY Index strength.

Correlation Dynamics: Gold remains a primary anchor, but divergence during specific volatility windows offers tactical opportunities.

Volume Analysis: Liquidity surges often serve as leading indicators for breakouts, as visualized in the multi-panel volume charts.

9. Conclusion and Future Scope

The project successfully demonstrates the efficacy of combining ensemble ML models with interactive BI dashboards. Future improvements will include live API data ingestion for real-time forecasting and the integration of natural language processing (NLP) for sentiment analysis of news sources.

10. References

- Historical Market OHLCV Datasets.
- Macroeconomic and Futures Database.
- Power BI and Python (Scikit-Learn/XGBoost) Documentation.